Youngwoo **Jeong**

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Education

New York University *New York, USA*

PH.D STUDENT IN ELECTRICAL AND COMPUTER ENGINEERING *Aug 2026 - Present*

- Advisor: Prof. Austin Rovinski
- Lab: REALISE Lab

Seoul National University of Science and Technology *Seoul, Republic of Korea*

M.S IN ELECTRONIC ENGINEERING *Mar 2022 - Feb 2024*

- Advisor: Prof. Seung Eun Lee
- Lab: Computer Architecture Laboratory
- Thesis: Approximate Arithmetic Circuits for Embedded Fuzzy Logic Controller

Seoul National University of Science and Technology *Seoul, Republic of Korea*

B.S IN ELECTRONIC ENGINEERING *Mar 2015 - Feb 2022*

- Advisor: Prof. Seung Eun Lee

Work Experience

MangoBoost *Seoul, South Korea*

SOFTWARE ENGINEER *Jan 2026 - July 2026*

- Developed a Kubernetes-native storage orchestration platform (Go Operator, Helm, React dashboard) managing DAOS distributed storage on bare-metal clusters.
- Implemented node lifecycle automation including NVMe/SPDK device management, SCM memory sizing, and multi-engine hugepage allocation.
- Operated and deployed multi-node K8s clusters with end-to-end provisioning pipeline (kubespray, Docker, Helm).

SYSTEM ARCHITECT *Mar 2024 - Dec 2025*

- Architected an NVMe/TCP initiator (NTI) with dynamic ARM–FPGA acceleration, allowing protocol processing to be flexibly offloaded depending on performance requirements and hardware availability. This ensured both efficiency and reliability in large-scale deployments (e.g., Ceph).
- Extended scalability from the original 2 subsystems to 18 FPGA-accelerated functions via SR-IOV, while achieving full 200G line-rate throughput in aggregated performance tests. On the ARM side, designed a flexible execution path supporting up to 130 concurrent functions to enable diverse application workloads.
- Verified functionality and throughput using FIO benchmarks across multiple configurations (single disk, RAID, XFS).
- Measured real-world database performance with PostgreSQL (pgbench) workloads.

Selected Publications [Full list on Google Scholar]

- **BoostX™-NTI: Fast, Scalable and Flexible Storage Architecture with NVMe/TCP Initiator Acceleration** [URL]
  The 53rd International Symposium on Computer Architecture (ISCA) Industry Track, 2026
  Hamin Jang, Jinha Jeong, Wonseok Lee, Jun Heo, Jongcheon Lee, Hyunjae Chu, Taehyun Kim, **Youngwoo Jeong**, Dongu Kim, Heetaek Jeong, Changsu Kim, Dongup Kwon, Jangwoo Kim
- **SEAM: A synergetic energy-efficient approximate multiplier for application demanding substantial computational resources** [URL]
  Integration. vol.101, 2025
  Youngwoo Jeong, Joungmin Park, Raehyeong Kim, Seung Eun Lee
- **Robot-Specific Processor for Autonomous Driving**
  1st Workshop on Robotics Acceleration with Computing Hardware (RoboARCH) (Co-located with MICRO), Chicago, USA, Oct., 2022
  Youngwoo Jeong, Kwang Hyun Go, Soohee Kim, Seung Eun Lee
- **An Architecture for Resilient Federated Learning through Parallel Recognition (Poster Session)** [URL]
  The 31st International Conference on Parallel Architectures and Compilation Techniques (PACT), Chicago, USA, Oct., 2022
  Jeongeun Kim, **Youngwoo Jeong**, Suyeon Jang, Seung Eun Lee
- **Photoplethysmography-Based Distance Estimation for True Wireless Stereo** [URL]
  Micromachines. vol.14, no.2, 2023
  Youngwoo Jeong, Joungmin Park, Sun Beom Kwon, Seung Eun Lee

Patents

Federated Learning Method and System Using Shared Learning Data *United States*

SEUNG EUN LEE, JEONGEUN KIM, **YOUNGWOO JEONG** *December 2023*

patent application

Method and System for Determining Final Result Using Federated Learning *United States*

SEUNG EUN LEE, JEONGEUN KIM, **YOUNGWOO JEONG** *December 2023*

patent application

Research Project

Development for Processing Software on AI Semiconductor Devices
South Korea

Ministry of Science and ICT — *2024 - 2022*

- Analyzed various AI models to standardize the input for AI systems.
- Proposed an architecture for a hardware scheduler optimized for multi-AI core architecture.

Development of Proximity/Healthcare Convergence Sensor SoC for TWS
South Korea

Ministry of Trade, Industry and Energy — *2023 - 2021*

- Designed a test environment for photoplethysmography sensors to evaluate their performance.
- Proposed a waveform adjustment filter and AI-based distance estimation algorithm to enhance signal processing accuracy.

Embedded AI Module Based on Neuromorphic Computing
South Korea

Ministry of Trade, Industry and Energy — *2021 - 2020*

- Designed various applications utilizing multiple embedded AI modules.
- Developed a testbed for evaluating multi-AI core controllers.
- Proposed methodologies to enhance accuracy in federated learning with multi-AI core systems.

Honors & Awards

School of Engineering Ph.D. Fellowship
New York, USA

New York University (NYU) — *August 2026*

Excellent Thesis Award
Seoul, South Korea

Seoul National University of Science and Technology — *February 2024*

- **Topic: Approximate Arithmetic Circuits for Embedded Fuzzy Logic Controller**

Corporate (LX Semicon) Special Award
Seoul, South Korea

Korea Semiconductor Industry Association — *October 2022*

- **Topic: AI Processor employing Stochastic Computing for Embedded Systems**

Department Chair Award
Seoul, South Korea

Seoul National University of Science and Technology — *February 2022*

- **Topic: Design of an Autonomous Indoor Robot for Robot-on-Chip**

Corporate (Silicon Mitus) Special Award
Seoul, South Korea

Korea Semiconductor Industry Association — *November 2021*

- **Topic: In-Vehicle Network Processor based on LIN and CAN-FD Controller**

Teaching Experience

Advanced AI Processor, Computer Architecture
Seoul, South Korea

Seoul National University of Science and Technology — *Fall 2022*

Teaching Assistant

Digital System Design, Resilient Processor Design
Seoul, South Korea

Seoul National University of Science and Technology — *Spring 2022*

Teaching Assistant

Skills

Hardware Description Languages	Verilog, SystemVerilog
High-Level Computer Languages	SystemC, C, C++, Python, Matlab
Design and Implementation Tools	Catapult HLS, Design Compiler, IC Compiler II, Quartus II, Vivado
Verification and Analysis Tools	Verdi, VCS, ModelSim, PSpice, PrimeTime, Formality, StarRC
Benchmark Tools	Flexible I/O (FIO), PGbench

Chip Design

Design of Robot-Specific Processor for Autonomous Driving

- Technology: Samsung 28nm RFCMOS
- Designer: **Youngwoo Jeong**, Yue Ri Jeong, Hyun Woo Oh, Kwang Hyun Go
- Gate Counts: 1062K @ 50MHz
- Memory: Code region (16KB), Data region (128KB)
- Date: 2022. 07. 18